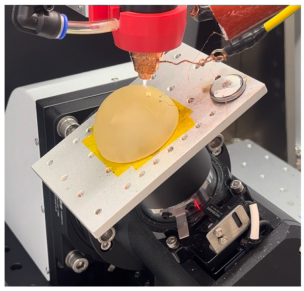
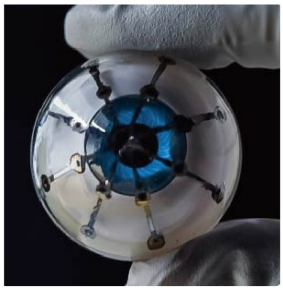
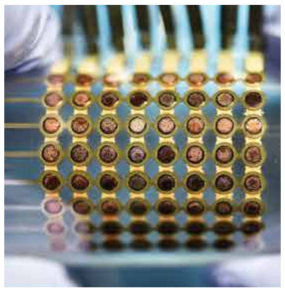
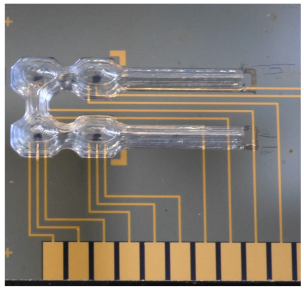
Ruitao Su

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My research group pioneers the field of 3D printed flexible and bio-integrated electronics with *in situ* nonthermal plasma activation to achieve low temperature, biocompatibility and high performance without any post treatment. We have established the on-the-fly interaction pathways between plasma reactive species and aerosolized precursors by experiments and simulations. The efficient material conversion and functionalization of conductors and semiconductors enable automated and conformal printing of 3D-structured bioelectronics such as bionic eyes and brain computer interfaces.

Professional Experience

Yongjiang Lab		
AFAM Group	Group Leader, Research Scientist	2024 – Now
Zhengzhou University		
School of Mech. and Power Eng.	Associate Professor	2022 – 2024
Massachusetts Institute of Technology		
CSAIL	Postdoctoral Associate	2021 – 2022
Advisor: Wojciech Matusik		
University of Minnesota		
Department of Mechanical Engineering	Postdoctoral Associate	2020 – 2021
Advisor: Michael C. McAlpine		

Education

University of Minnesota	Ph.D. in Mechanical Engineering	Oct. 2020
Advisor: Michael C. McAlpine		
Best Dissertation Award: 3D Printing Multifunctional Optoelectronic and Microfluidic Devices		
University of Cincinnati	M.S. in Mechanical Engineering	July 2015
Advisor: Mark J. Schulz		
Thesis: Multi-Field Physics for the Synthesis of Carbon Nanotube Yarn and Sheet		
Huazhong University of Sci. and Tech.	B.S. in Mechanical Engineering	July 2013

Grants

- Property Control and Mechanisms of High-Resistivity Soft Magnetic High-Entropy Alloys Fabricated by LPBF. NSFC Youth Program, Research-group-led project, 2025-2027, RMB 300,000.

- Mechanisms of 3D Printing Self-Supporting Microscale Thin-Shell Structures Using the Yield Behavior of Thixotropic Fluids. NSFC Youth Program, Principal Investigator, 2024-2026, RMB 300,000.
- Mass and Energy Transport Mechanisms in Low-Temperature Plasma Jet 3D Printing of Functional Materials for Microelectronics Manufacturing. Zhejiang Provincial Natural Science Foundation, Key Program, Principal Investigator, 2026-2028, RMB 300,000.
- In Situ Plasma Reduction of Precursors and Structure Formation Mechanisms for Low-Temperature 3D Printing of Conductive Metals. China Postdoctoral Science Foundation, General Program, Research-group-led project, 2027-2029, RMB 160,000.
- Formation Mechanisms of 3D Printed Self-Supporting Microscale Thin-Shell Structures for Electronic Device Packaging. China Postdoctoral Science Foundation, General Program, Principal Investigator, 2024-2025, RMB 160,000.
- CAD Cloud Architecture for Collaborative Product Design in Ubiquitous Computing Environments. National Key R&D Program of China, Young Scientists Project, Project participant, 2023-2026, RMB 2,000,000.
- Early Diagnosis and Risk Stratification of Prostate Cancer Using Nanozyme-Enhanced Multitarget Electrochemical Sensing. Zhejiang Provincial Natural Science Foundation, Exploratory Program, Project participant, 2026-2027, RMB 100,000.

Publications

Manuscripts Under Revision

- B. An, X. Guo, X. Chen, **R. Su**, Low-Temperature 3D Printing of Silver Conductors via Rapid On-the-Fly Synthesis in a Nonthermal Plasma-Aerosol Jet, *Advanced Materials* (Under Revision)
- **R. Su**, X. Chen, Quantifying the Self-Supporting Geometry Limits of Direct Ink Writing: Rheology-Based Design Rules for a Viscoelastic Silicone Ink, *Materials & Design* (Under Revision)
- **R. Su**, X. Chen, The Effect of Structural Evolution on the Electrical Resistivity of Binder Jet 3D Printed Porous Silicon Carbide Ceramics, *Additive Manufacturing Frontiers* (Under Revision)

Peer-Reviewed Articles

- **R. Su**, F. Wang, M. C. McAlpine, 3D Printed Microfluidics: Advances in Strategies, Integration, and Applications, *Lab on a Chip* **23***,* 1279-1299 (2023)
- **R. Su**, S. H. Park, X. Ouyang, S. I. Ahn, M. C. McAlpine, 3D Printed Flexible Organic Light-Emitting Diode Displays, *Science Advances* **8**, eabl8798 (2022)
  - Highlighted: *Nature* (2022). DOI: 10.1038/d41586-022-00043-4
- X. Ouyang, **R. Su**, G. Han, D. W. H. Ng, D. R. Pearson, M. C. McAlpine, 3D Printed Skin-Interfaced UV-Visible Photodetectors, *Advanced Science* **9***,* 2201275 (2022)
- **R. Su**, J. Wen, Q. Su, M. S. Wiederoder, S. J. Koester, J. R. Uzarski, M. C. McAlpine, 3D Printed Self-Supporting Elastomeric Structures for Multifunctional Microfluidics, *Science Advances* **6**, eabc9846 (2020)
- S. H. Park* (co-first), **R. Su* (co-first)**, J. Jeong, S.-Z. Guo, K. Qiu, D. Joung, F. Meng, M. C. McAlpine, 3D Printed Polymer Photodetectors. *Advanced Materials* **30**, 1803980 (2018)
  - Highlighted: *Nature* (2018). DOI: 10.1038/d41586-018-06193-8
- K. Qiu, Z. Zhao, G. Haghiashtiani, S.-Z. Guo, M. He, **R. Su**, Z. Zhu, D. Bhuiyan, P. Murugan, F. Meng, S. H. Park, C.- C. Chu, B. M. Ogle, D. A. Saltzman, B. R. Konety, R. M. Sweet, M. C. McAlpine, 3D Printed Organ Models with Physical Properties of Tissue and Integrated Sensors. *Advanced Materials Technologies* **3**, 1700235 (2017)
- G. Hou, D. Chauhan, V. Ng, C. Xu, Z. Yin, M. Paine, **R. Su**, V. Shanov, D. Mast, M. Schulz, Y. Liu, Gas Phase Pyrolysis Synthesis of *Carbon* Nanotubes at High Temperature. *Materials & Design* **132**, 112-118 (2017)

- G. Hou, **R. Su**, A. Wang, V. Ng, W. Li, Y. Song, L. Zhang, M. Sundaram, V. Shanov, D. Mast, D. Lashmore, M. Mark, Y. Liu, The effect of a convection vortex on sock formation in the floating catalyst method for carbon nanotube synthesis. *Carbon* **102**, 513–519, (2016)

Book Chapter

- **R. Su**, S. H. Park, Z. Li, M. C. McAlpine, "3D Printed Electronic Materials and Devices," in Robotic Systems and Autonomous Platforms: Advances in Materials and Manufacturing. Eds: S. M. Walsh, M. S. Strano. CH 13 (Woodhead, Cambridge, 2019)

Conference Proceeding

- J. R. Uzarski, M. S. Wiederoder, C. Luckhardt, R. Paffenroth, **R. Su**, M. C. McAlpine, Novel data science driven chemical and biological agent sensors: towards better discrimination in complex environments, *18th International Meeting on Chemical Sensors*, Montreal, Canada (2020)

Patents

Granted Chinese Patents

- **R. Su**, H. Tu, B. An, X. Zhou, X. Chen, "PEDOT:PSS Fabrication Using a Plasma-Aerosol Hybrid Jet, Printed Structures, and Applications," Chinese Patent ZL 2025 1 2042398.6, 2026.
- **R. Su**, B. An, H. Tu, X. Zhou, X. Chen, "3D Printing Nozzle, System, and Low-Temperature Plasma Jet 3D Printing Apparatus," Chinese Patent ZL 2025 1 1917097.7, 2025.
- **R. Su**, H. Tu, X. Zhou, K. Zheng, "Binder Jet Printing Method, 3D Printed Components, and Binder Jet Printing Apparatus," Chinese Patent ZL 2026 1 0699462.X, 2026.

U.S. Patents and Provisional Application

- M. C. McAlpine, **R. Su**, S. J. Koester, J. R. Uzarski, "Additively Manufactured Self-Supporting Microfluidics," U.S. Patent US 11,820,061 B2.
- M. C. McAlpine, **R. Su**, S. H. Park, "Organic Light-Emitting Diode (OLED) Display and Methods of Fabrication Using a Multimodal Three-Dimensional (3D) Printing Technique," U.S. Provisional Patent Application 63/247,358.
- M. C. McAlpine, X. Ouyang, D. Pearson, **R. Su**, "Photodetectors for Measuring Real-Time Optical Irradiance," U.S. Patent US 12,723,913 B2.

Chinese Patent Applications

- **R. Su**, X. Chen, W. Chen, H. Li, "Additive Manufacturing Apparatus with an Integrated In Situ Plasma Energy Source and Manufacturing Method," Chinese Patent Application CN 118810023 A, 2024.
- **R. Su**, W. Wang, H. Tu, B. An, X. Chen, "Low-Temperature Plasma-Aerosol Hybrid Jet Nozzle and 3D Printing Equipment," Chinese Patent Application 202511251469.7, 2025.
- **R. Su**, B. An, X. Ren, "Chip Assembly and Preparation Method," Chinese Patent Application CN 121136817 A, 2025.
- **R. Su**, B. An, X. Lu, "Low-Temperature Plasma Jet 3D Printing Method, Printed Structures, and Applications," Chinese Patent Application 202610018862X, 2025.
- **R. Su**, X. Zhou, H. Tu, B. An, G. Yang, "3D Printed Silicon Carbide Ceramics, Preparation Method, and Products," Chinese Patent Application CN 122325233 A, 2026.
- **R. Su**, B. An, X. Lu, "Low-Temperature Plasma Jet 3D Printing System and Apparatus," Chinese Patent Application 202611501219.9, 2026.

U.S. and International Patent Applications

- M. C. McAlpine, X. Ouyang, D. Pearson, **R. Su**, "Photodetectors for Measuring Real-Time Optical Irradiance," International Patent Application WO2023/244975A2.

- M. C. McAlpine, **R. Su**, S. H. Park, "Organic Light-Emitting Diode (OLED) Display and Methods of Fabrication Using a Multimodal Three-Dimensional (3D) Printing Technique," International Patent Application PCT/US22/44322.
- M. C. McAlpine, **R. Su**, S. J. Koester, J. R. Uzarski, "Additively Manufactured Self-Supporting Microfluidics," International Patent Application PCT/US2020/061072.
- E. Crist, D. K. Wood, **R. Su**, M. C. McAlpine, "Three-Dimensional Microfluidic Metastasis Array," U.S. Patent Application US 2024/0189822 A1. International Patent Application PCT/US22/71843.

Presentations

Invited Talks and Seminars

- "Novel Electronic Structures and Devices via Low-Temperature Plasma Jet 3D Printing," National Forum on Advanced Electronic Information Materials and Intelligent Sensing Technologies, Suzhou, China, 2026.
- "3D Printing Functional Materials and Devices," 5th Additive Manufacturing Innovation Conference, Ningbo, China, 2026.
- "Key Roles of Nonthermal Plasma and Other Novel Energy Sources in 3D Printing," 4th Summit on Polymer 3D Printing Materials, Hangzhou, China, 2025.
- "3D Printing Multifunctional Microfluidics," Washington State University, Online, 2022.
- "3D Printing Optoelectronic Materials and Devices," KLA Instruments Display Materials Technology Asia Symposium, Online, 2022.
- "3D Printed Microfluidics with Applications in Drug Screening and Oncology Research," 6th Annual 3D Tissue Models Summit, Boston, MA, USA, 2021.
- "3D Printed Self-Supporting Elastomeric Microfluidics with Yield-Stress Polymers," Korea Institute of Industrial Technology, Online, 2021.
- "3D Printing Functional Materials for the Next-Generation Optoelectronic & Microfluidic Devices," 9th International Young Scholars Donghu Forum, Huazhong University of Science and Technology, Online, 2020.

Conference Presentations

- "3D Printed Flexible Organic Optoelectronic Arrays," LOPEC 2023 Scientific Conference, Munich, Germany, 2023.
- "3D Printed Flexible Organic Light Emitting Diode Displays," Materials Research Society Meeting, Boston, MA, USA, 2021.
- "3D Printed Multifunctional Devices for Biomedical Applications," Materials Research Society Meeting, Online, 2020.
- "3D Printed Self-Supporting Elastomeric Structures for Multifunctional Microfluidics," Materials Research Society Meeting, Online, 2020.
- "3D Printed Polymer Photodetector," Materials Research Society Meeting, Boston, MA, USA, 2018.

Posters

- "3D Printed LED and Photodetectors" July 2019
  Poster on Purdue Mi-Bio Summit on Flexible and Stretchable Bioelectronics (West Lafayette, IN)
- "3D Printed Silicon Nanocrystal LED" Mar. 2017
  Poster on NSF MnDRIVE Symposium (Minneapolis, MN)

Honors & Awards

Ningbo Leading Talent 2025

Ningbo Municipal Human Resources and Social Security Bureau

Zhejiang Provincial High-Level Talent 2025
Department of Human Resources and Social Security of Zhejiang Province
Pui Best Dissertation Award 2022
University of Minnesota
Best Dissertation Award 2021
Department of Mechanical Engineering of UMN
MRS Best Presentation Award 2020
Material Research Society, Additive Manufacturing Symposium
MRS Graduate Student Silver Award 2018
Material Research Society (https://www.mrs.org/gsa-past)
Outstanding Research Award 2015
Nanoworld Lab at the University of Cincinnati
National Encouragement Scholarship 2010
Ministry of Education of the People's Republic of China
Excellent Freshman Study Scholarship 2010
Mechanical School of HUST

Academic Services

- **Journal topic editor**: *Micromachines* (since 2021)
- **Journal Youth Editorial Committee:** *Additive Manufacturing Frontiers* (since 2023)
- **Journal reviewer**: *Nature Communications, Communications Engineering, MRS Advances, Journal of Materials Chemistry C, ACS Applied Materials & Interfaces, npj Flexible Electronics, PLOS ONE*

Teaching & Services

- **Guest lecturer**: Introduction to Nanoscience (2021 Fall, Virginia Commonwealth University)**,** Biofabrication (2021 Spring, University of Nebraska-Lincoln)**,** Introduction to Nanotechnology (2019 Fall, University of Minnesota)
- **Teaching assistant**: Vibration Engineering (2016 Spring, University of Minnesota), Kinematics and Kinetics of Machines, Structural Mechanics, Engineering Economics (2013 Fall, University of Cincinnati)
- **Lab safety officer** of the McAlpine Research Lab at University of Minnesota (2017 – 2020)
- **Secretary** of ME Grad Student Council, University of Minnesota (2017 – 2019)
- **Volunteer supervisor** of environmental protection nonprofit – Great River Greening (2017)

Group Members & Alumni

Postdoctoral Researchers

- **Bailing An:** Current postdoctoral researcher; Principal Investigator, NSFC Youth Program.
- **Kaihang Jin:** Former postdoctoral researcher; currently Lecturer at Zhejiang Wanli University.
- **Xin Zhou:** Former postdoctoral researcher; currently a postdoctoral researcher in Sweden.

Doctoral Students

- **Saba Mubashar:** Current Ph.D. student, joint training: University of Nottingham Ningbo China / Yongjiang Lab.
- **Naeem Ashraf:** Current Ph.D. student, joint training: University of Nottingham Ningbo China / Yongjiang Lab.
- **Xiaopei Ren:** Current Ph.D. student, joint training: University of Nottingham Ningbo China / Yongjiang Lab.

Research Interns and Alumni

- **Wentao Wang:** Currently pursuing a Ph.D. at The Chinese University of Hong Kong.

- **Nicholas Fuhr:** 2017 NSF MRSEC summer undergraduate researcher; Ph.D., Boston University; currently Principal Scientist at Physical Sciences, USA.
- **Zijian Guo:** Currently pursuing graduate studies at German RWTH Aachen University.
- **Yuhua Hu:** Currently pursuing graduate studies at Southern University of Science and Technology.

Engineers

- **Xiaoting Lu:** Plans to apply for Ph.D. study.
- **Haowei Tu:** Currently an engineer at Ningbo Additive Manufacturing Center.

Press Report

- "Print job completed: a bendable image display," *Nature Research Highlight*, Jan. 2022
- "How microfluidics can automate drug discovery and development," *Drug Target Review*, May 2021
- "3D printing microfluidic channels for medical testing," *National Academy of Engineering Frontier of Engineering*, Jan. 2021
- "Eyes, wasps and asteroid dust — August's best science images," *Nature News*, Sep. 2018
- "12 innovations that will revolutionize the future of medicine," *National Geographic*, Dec. 2018